App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a Course Instructor at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

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2. Project Details

Project Title: Quantified Self App

Problem Statement:

To design and build a web-based application that allows users to **track, visualize, and reflect on their daily habits and activities**, such as sleep, exercise, study hours, and mood.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows users to log activities daily, visualize trends through interactive charts, and analyze behavioral insights over time.

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3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, creating API documentation samples, and improving variable naming consistency.
The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and documentation formatting**.
All final implementation logic, debugging, and integration were done manually.

OR

No use of AI/LLM.

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4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Chart.js	To visualize trends such as activity vs. time
Flask-Login	User authentication and session management
WTForms	Frontend form validation
SQLite	Lightweight local database for storing user data

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5. Database Schema / ER Diagram

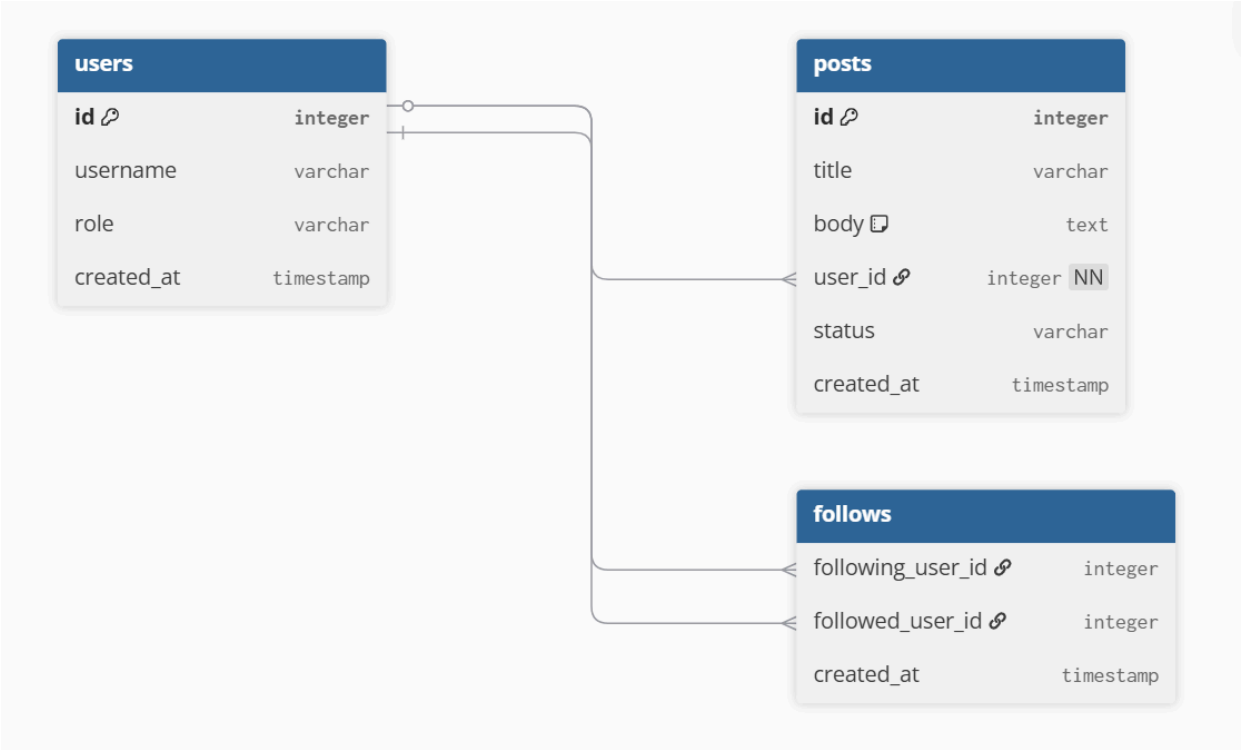
Tables:

1. **User** — stores user profile details (id, name, email, password)
2. **Activity** — logs user activities (id, user_id, activity_type, duration, date, mood)
3. **ActivityType** — stores predefined activity categories (id, name, description)

Relationships:

- One-to-Many → `User → Activity`
- One-to-Many → `ActivityType → Activity`

(Insert ER diagram image here – can be created using dbdiagram.io or draw.io or any other website)



(example taken form dbdiagram.io)

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6. API Resource Endpoints

Endpoint	Method	Description
`/api/login`	POST	Authenticate user and generate session
`/api/activities`	GET	Fetch logged activities for a user
`/api/activity/add`	POST	Add new activity entry
`/api/summary`	GET	Return daily/weekly analytics summary

YAML API Definition File:
Included separately in the submission ZIP as `api.yaml`.

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7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for user and activity routes
- **/templates** – Jinja2 HTML templates
- **/static** – CSS, JS, and chart visualization files

Implemented Features:

User registration and login
Daily activity tracking and habit logging
Visualization dashboard using Chart.js
Summary analytics with date filtering
Responsive UI built with Bootstrap
Secure password hashing and user session handling

Additional Features:

- Export user logs as CSV
- AI-generated weekly summary insights (optional extension)

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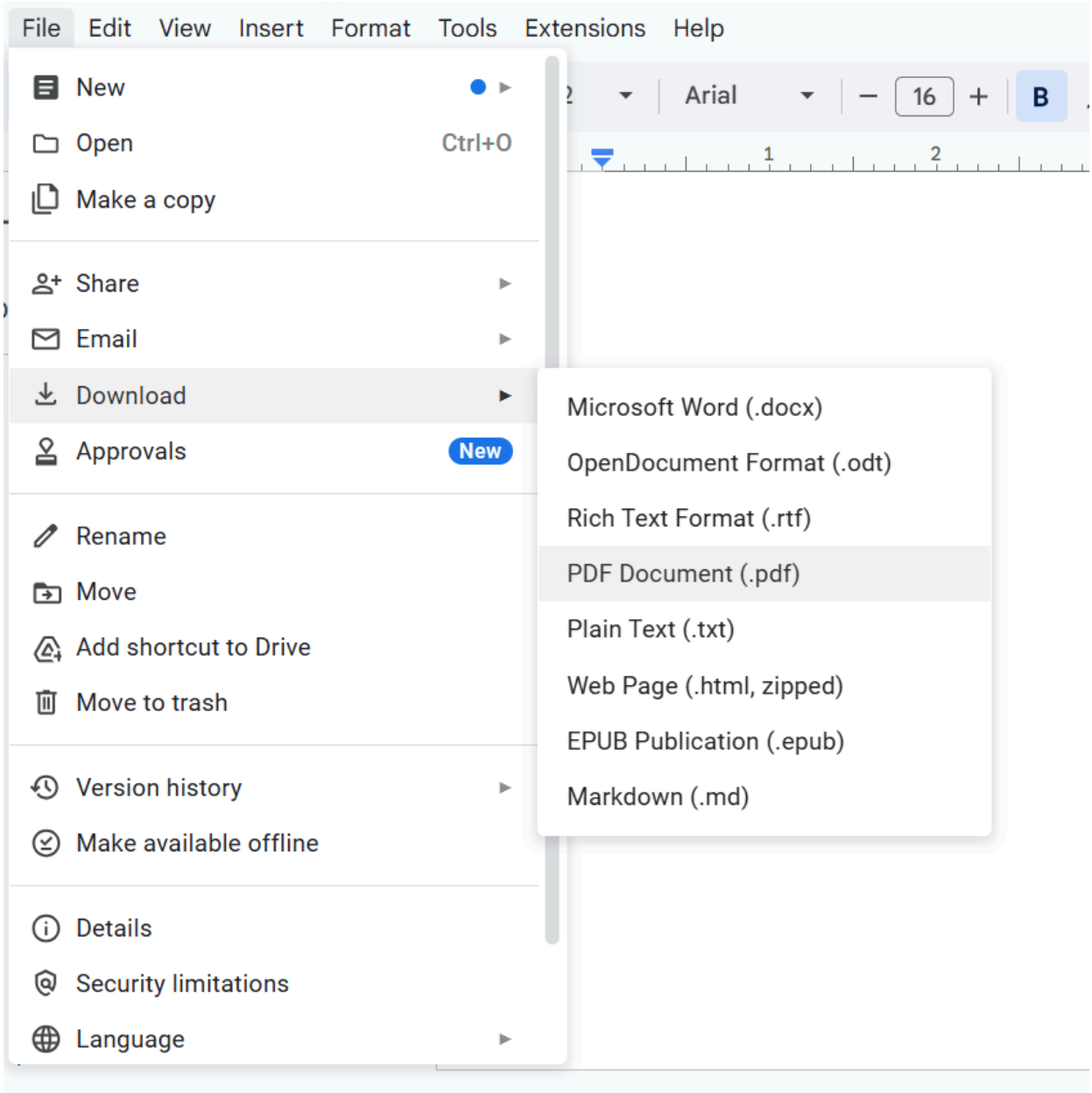
8. Video Presentation

Drive Link:
https://drive.google.com/file/d/your-video-link-here
(Accessible to all with "View" permission.)

After Creating the doc

Convert this doc to pdf and download (This should not be included in project report.):



Where should you be adding the report ?

- The report should be included in the zip file which you are submitting.
- Put the report in the drive link and paste the drive link in the viva portal while submitting the app.

Both the things are covered in the video. Please watch it once.

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